Software Architecture Specification

i3 AI-Lab Assessment & Certification Engine

Doc ID: ARCH-I3-LABS-2026-V1	**Status:** Approved	**Target Engine:** Cloud-Native Microservices

1. Platform Overview

The i3 AI-Lab Assessment Engine evaluates real-world coding capability, automates admissions grading, and runs enterprise-grade certification drills inside sandboxed microenvironments orchestrated by autonomous AI evaluation agents.

2. System Components

Service Name	Runtime Stack	Primary Responsibility
`svc-admissions-hook`	Node.js / Express	Candidate ingestion, lifecycle callbacks, and CRM sync.
`svc-assessment-catalog`	Go (Golang)	Challenge specs, blueprints, dynamic prompt seeds.
`svc-sandbox-orchestrator`	Go / Nomad / K8s	Provisions isolated, time-boxed Docker execution pods.
`svc-agent-evaluator`	Python / LangGraph	Multi-agent evaluation (Architect, Adversarial QA, Copilot Audit).
`svc-certification-reports`	Node.js / PostgreSQL	Aggregates grades, renders badges, and fires webhook responses.

3. Database Schema Definitions

```sql
CREATE TABLE assessment_sessions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    admissions_applicant_id VARCHAR(128) NOT NULL,
    blueprint_id UUID REFERENCES assessment_blueprints(id),
    status VARCHAR(32) NOT NULL DEFAULT 'PENDING',
    sandbox_endpoint VARCHAR(255),
    session_started_at TIMESTAMPTZ,
    expires_at TIMESTAMPTZ NOT NULL
);
```

4. Webhook Callback Specification

```
POST /api/webhooks/candidate-result HTTP/1.1
Content-Type: application/json
X-i3-Signature: sha256=...

{
  "applicant_id": "APP-2026-9812",
  "passed": true,
  "total_score": 88.5,
  "performance_band": "PRODUCTION_READY",
  "recommendation": "FAST_TRACK_ADMIT"
}
```